



Emulium[®] Mellifera MB

Designed to delight



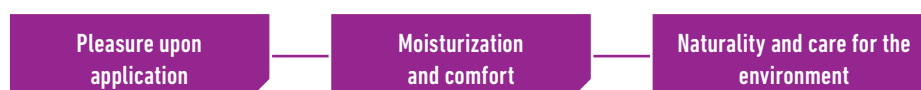
People make our name



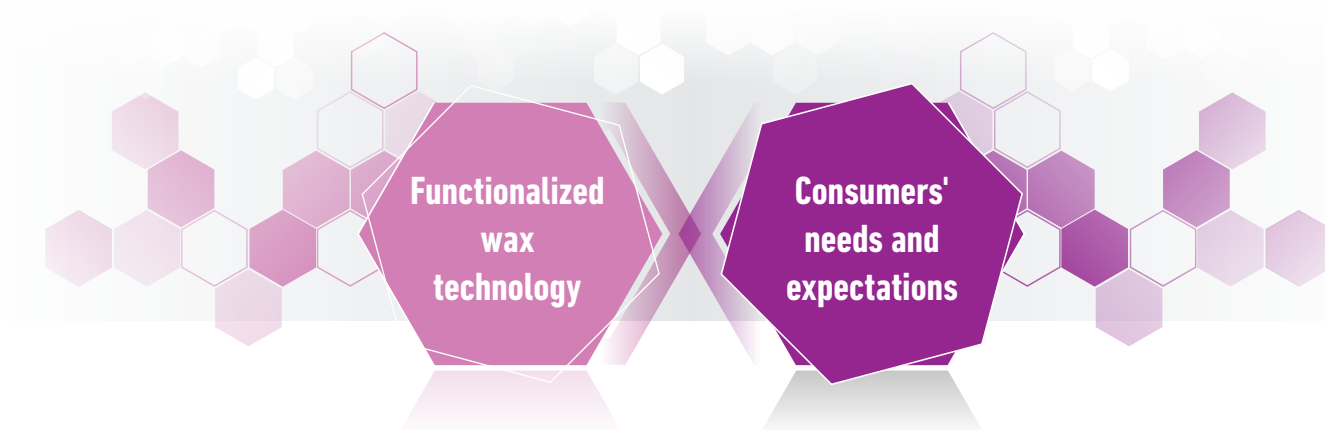
Flexible & multifunctional emulsifier

From light milky lotions to rich butters, **our preferences for texture vary considerably** and are strongly influenced by **skin type, climate** and **cultural differences**. Across the world, environmental conditions play a significant role on perception of sensory properties. Cold winters, or hot and humid summers, the needs of the skin vary and require a large choice of cosmetic textures.

However, despite climate differences, there are common requirements across the planet:

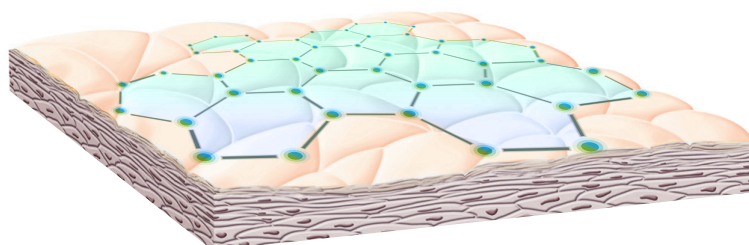


Recognizing the need for **flexible ingredients** that meet the expectations of worldwide consumers, Gattefossé has developed an innovative sensorial emulsifier. Based on a patented technology which transforms and functionalizes natural waxes, this ingredient **auto-adapts** to the environment in which we live, bringing guaranteed pleasure to all users.



EMULIUM® MELLIFERA MB, *the first emulsifier designed to meet the expectations of all cosmetic users*

Emulium® Mellifera MB forms a **soft** and **ultra-light film** on the skin's surface, providing a pleasurable touch and protection from seasonal variations.



Emulium® Mellifera MB is created from natural ingredients using green chemistry. Beeswax and jojoba wax are reacted with vegetable polyglycerol to change the texture into a soft and homogeneous butter that possesses active properties. This functionalized wax butter is then combined with polyglyceryl-6 distearate and cetyl alcohol, in a precise composition designed to optimize emulsification and texturizing capacity.

Sensoriality and activity

IN VIVO
TESTED

Fine and light sensation on the skin

Emulium® Mellifera MB creates textures of great beauty and fineness, characterized by a **white** and **glossy luxurious aspect**. The sensation on the skin is **light and smooth**, providing a **barely perceptible residual film** that drapes the skin with comfort. ⁽¹⁾



Auto-adaptation to climate

Emulium® Mellifera MB is the first ingredient to **auto-adapt to the environment**.

- In cold or dry conditions, it creates a soft, comfortable film of protection on the skin.
- In hot and humid conditions, Emulium® Mellifera MB **stabilizes the sensory properties**, leaving the skin less heavy and less greasy with a lower perceptible film. ⁽²⁾



Application delight

The textures formulated with Emulium® Mellifera MB bring pleasure upon application **whatever the skin type**.

After the first application, the **fresh sensation** and **easy absorption** were noted by the panelists. **100% of volunteers said the product suits their skin**.

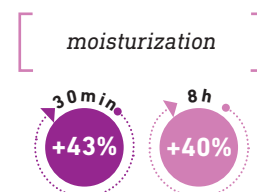
After 3 weeks of use, the textures continued to delight all volunteers. ⁽³⁾

Skin
adaptation
100%

Long-lasting moisturization

Emulium® Mellifera MB brings intense **immediate and long-lasting moisturization** to the skin: +43% after 30 minutes, +40% after 8 hours. ⁽⁴⁾

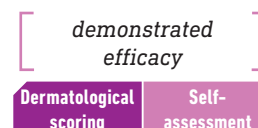
After only one application, 100% of panelists confirm their **skin is visibly better hydrated** ⁽³⁾.



Visible improvement of skin appearance

When the skin feels good, the results are visible and perceived by the panelists. Confirmed by a dermatologist, Emulium® Mellifera MB brings highly **significant improvement to the quality of the skin**.

A formulation with Emulium® Mellifera MB gave a **77% increase in skin radiance, +68% skin texture fineness, +58% smoothness, +35% skin tonicity aspect**, visibly replumped and moisturized skin. ⁽⁵⁾



⁽¹⁾ Sensorial study carried out by a sensorial panel on 20 textures

⁽²⁾ Sensory analysis, in tropical conditions, of a cream with 1% Emulium® Mellifera MB wax butter vs placebo

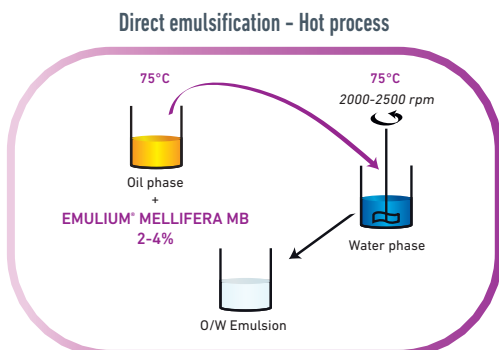
⁽³⁾ Use test on 21 volunteers with very dry to oily skin (22-50 years old) - Applications twice a day on the face during 21 days - Emulsion with 4% Emulium® Mellifera MB

⁽⁴⁾ Corneometry test on 10 volunteers with dry skin (22-68 years old) - Measurements at T 30 min, T 4 H, T 8 H vs T 0. Single application on the forearm area - Emulsion with 4% Emulium® Mellifera MB vs non treated area

⁽⁵⁾ Dermatological scoring and self-assessment on 21 Asian volunteers with very dry to oily skin (22-50 years old) - Applications twice a day on the face during 21 days - Emulium® Mellifera MB at 4% in a simple emulsion containing no active ingredient

Formulation and applications

Easy to formulate



- 1 Add Emulium® Mellifera MB to the oil phase and heat to **75°C**
- 2 Once homogeneous, **slowly add the oil phase** to the heated aqueous phase under high shear (using rotor stator or dispersing mixers)
- 3 **Maintain high shear mixing** until the emulsion is homogeneous (approx. 10-20 minutes)
- 4 Cool under mixing

The use of a gelling agent is recommended to help the stabilization of the emulsion, particularly at high temperatures (ex: xanthan gum 0.2% + cellulose, other gums or polymers)

Wide compatibility

Oils	All types up to 35%
Consistency agents	Most consistency agents (max. 1% of glyceryl stearate)
Gelling agents	Most gelling agents (max 0.2% of acrylate polymers and copolymers that are not pre-neutralized)
Actives	Most actives at a pH between 4-10
Mineral and organic UV filters	Use of a co-emulsifier recommended (anionic emulsifiers such as potassium cetyl phosphate at 0.2%)
Pigments	All types of pigments
Electrolytes	Up to 2% NaCl in presence of a gelling agent
Alcohol	Up to 10%
Preservatives	Most preservative systems (If using acidic preservatives, it is recommended to add them after emulsification)

A wide range of applications

Natural & certifiable formulations

100% natural, ECOCERT ⁽⁶⁾ & COSMOS ⁽⁷⁾ compliant, NPA™ certified, sustainable sourcing

Fully flexible skincare textures

From sprayable lotions to thick butter creams; particularly well suited to low viscosity formulations

Second skin color cosmetics

Fine emulsions, low residual film, excellent transfer resistance

Highly sensorial sun care

Beautiful fluid emulsions, non-oily, non-sticky with a light after-feel



Emulium[®] Mellifera MB

brings a sensory solution to global consumer expectations, bringing delight upon application and visible efficacy.

Created and formulated by teams across the world, Emulium[®] Mellifera MB allows the creation of **stunning, luxurious textures that bring **pleasure** and **activity**.**

- Allows the formulation of a wide range of textures suitable for all skin types and climates
- Creates natural emulsions with white and luxurious appearance
- Shows significant moisturizing properties and visibly improves skin texture

Draped in a soft and barely perceptible protective veil, the skin, protected from seasonal extremes, is visibly more radiant and beautiful.

Characteristics

- O/W emulsifier (pellets)
- PEG-free
- Non-ionic

Applications

- Skin care
- Color cosmetics
- Sun care
- Natural and organically certified formulations
- Adapted to sensitive and hyper reactive skin (clinically tested)

Sensoriality

- White and glossy textures
- Very low residual film
- Smooth and comfortable after-feel

INCI: Polyglyceryl-6 Distearate (and) Jojoba Esters (and) Polyglyceryl-3 Beeswax (and) Cetyl Alcohol

Worldwide approval for cosmetic applications

Listed on IECIC (Inventory of Existing Cosmetic Ingredients in China) 2003
ECOCERT⁽⁶⁾ and COSMOS⁽⁷⁾ compliant

⁽⁶⁾ The product has been certified by Ecocert Greenlife according to the Ecocert Natural and Organic Cosmetics Standard

⁽⁷⁾ The product has been validated by Ecocert Greenlife according to the COSMOS-standard of Organic and Natural cosmetics

The information given in this brochure is presented in good faith and we believe that it is correct at the moment of printing, but no warranty as to accuracy of results or fitness for a particular use is given. It cannot substitute for indispensable preliminary tests that must be conducted to ensure product suitability for each intended use. Nothing herein shall be construed as a recommendation or license to use any information that conflicts with any patent or trademark of our company or others. We make no representations or warranties, express or implied that any use of this information will not infringe any such patent or trademark. The user's attention is also drawn to the possible risks incurred if a product is used for a purpose other than its designed purpose. The user shall be solely liable for the precautions taken relevant to the use made of the products. This information does not exempt the user from ensuring that he may possibly come under other obligations pursuant to laws or regulations other than those indicated relating to the possession and manipulation of the product for which he shall be solely liable.

As a user of palm oil and palm kernel oil derivatives, Gattefossé is sensitive to the environmental impact of palm tree growing and participates in the development of sustainable cultivation, providing its customers with environmentally responsible ingredients.

Gattefossé is a proactive member of RSPO (Roundtable on Sustainable Palm Oil) since 2010. In 2013, Gattefossé achieved the RSPO supply chain certification. Since then, Gattefossé offers products certified RSPO Segregated, and more recently, products certified RSPO Mass Balance.

Emulium[®] Mellifera MB is certified Mass Balance.



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